

AMITH Civil & Allied Engineering Services - Website Documentation & Redesign Specification

This documentation acts as a complete blueprint of the original website (<https://amith.in.net/>) and outlines the structural contents, navbar details, copy, credentials, and contact info. It also provides a gap analysis and actionable recommendations for the Next.js Redesign Project (`AMITH_REDESIGN`).

1. Executive Summary & Core Identity

AMITH Civil & Allied Engineering Services Pvt Ltd (ACAES) is a specialized consulting and solution provider in the area of **Civil Structural & Construction Management**.

- Accredited Laboratory Association:** The company has a formal tie-up/MoU with **Tech-Civil Material Testing Laboratory Pvt Ltd (NABL Accredited)**, Perungalathur, Chennai, for construction material testing and Non-Destructive Testing (NDT) of concrete elements.
- Key Value Proposition:** Providing need-based technology, engineering advisory, and durable solutions for new construction as well as specialized, one-time repair and rehabilitation/retrofitting schemes for existing RCC structures and infrastructure (such as buildings, bridges, and RCC works).

2. Page Architecture & Global Details

2.1 Navigation Menu (Navbar) Mapping

Old Website Page	Old Filename / URL	Key Section / Component in Redesign
Home	<code>index.html</code> (https://amith.in.net/)	<code>#home</code> (Hero, Services preview)

About Us	non-destructive-and-partial-destructive-test-of-concrete-chennai.html	#about (WhyChooseUs component)
Services	quality-control-inspection-services-in-chennai.html	#services (Services component)
Gallery	repair-and-rehabilitation-of-concrete-structures-chennai.html	#projects / Gallery section
Contact Us	contact.html	#contact (ContactBanner, Footer)

2.2 Global Contact Information

- **Office Address:** No.35/F3, Sai Krupa Apartment, Ramagirinagar, Taramani Link Road, Velachery, Chennai - 600042.
- **Phone Number:** +91 9940548833
- **Official Email Addresses:**
 - Primary: amithcivilengineering@gmail.com
 - Sales: sales@amith.in.net
 - Admin: admin@amith.in.net
- **Social Platforms (Placeholders):** Facebook, Twitter, LinkedIn, Instagram.

3. Page-by-Page Content & Copy Extraction

3.1 Home Page

- **Meta Title:** Structural Audit Consultants Chennai, Quality Audit Services Chennai
- **Meta Description:** AMITH Civil & Allied Engineering provides Structural Audit Consultants Chennai and Quality Audit Services in Chennai. Analyse existing buildings, bridges & other RCC infrastructure works.
- **Core Banner Taglines:**
 - "Welcome to AMITH – Your Journey Begins Here"
 - "Build Your Quality With Our Service"

- **High-Level Offerings Previewed:**

- Testing of building materials
 - Restoration & rehabilitation of structures
 - Concrete mix design
 - Consultancy & Advisory services
 - Dissemination of technical knowledge
 - QA/QC & TPQM of civil projects
 - Quality Audit/ Assessment
 - Structural auditing
 - NDT & Partial destructive Testing
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3.2 About Us Page

- **Meta Title:** Non-Destructive and Partial Destructive Test of Concrete Chennai

- **Corporate Vision:**

"To provide need-based technology, Engineering & consultancy services for the construction industry with durable solutions for new construction & one-time solutions for repair & rehabilitation/retrofitting the concrete elements of existing buildings & infrastructure."

- **Corporate Mission:**

1. *Provide affordable solutions to the needs of stakeholders in concrete technology by integrating innovative technology for the construction industry.*
2. *Provide structural auditing of existing buildings, bridges & other RCC infrastructure works and appropriate end solutions to enhance the functionality, durability & service life of the structure by appropriate retrofitting techniques.*
3. *Provide the construction industry with appropriate & durable construction methodology for new construction.*
4. *Dissemination of knowledge by conducting a training program for Quality auditing/assessment, execution of repair, rehabilitation & retrofitting of concrete structural elements using appropriate methodology.*

- **Distress Assessment Philosophy:**

- Changes in building usage, load increases, and environmental exposures lead to structural distress. ACAES specializes in analyzing these distress indicators to formulate precise structural retrofitting specifications.
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3.3 Board of Directors (Complete Credentials)

The core strength of AMITH lies in the academic, operational, and administrative credentials of its board.

1. Professor Dr. Devadas Manoharan M

- **Background:** Academician & Administrator with **37+ years of experience**.
- **Roles Held:**
 - Former Vice Chancellor of *Anna University of Technology, Tiruchirappalli*.
 - Former Head of the Department of Civil Engineering, Anna University, Chennai.
- **Research & Guidance:** Guided 12 Ph.D. scholars, 91 M.E. theses, and 3 M.S. research projects.
- **Key Industrial Projects:**
 - Designed/vetted over 10 Sewage Treatment Plants (STPs) ranging from 8.5 MLD to 120 MLD capacities.
 - Supervised/vetted major infrastructure for *North Madras Thermal Power Plant, Airport Authority of India (runways), Tamil Nadu Slum Clearance Board (Tsunami-affected tenements), Bharat Petroleum Corporation Limited (BPCL), and Integral Coach Factory (ICF)*.

2. Col. Dr. Nallathambi P

- **Background:** Retired Army Officer from the **Military Engineering Services (MES)** (20+ years of active service) and Structural Consultant (10+ years of commercial practice).
- **Technical Expertise:** Specializes in Pre-Engineered Buildings (PEB) steel structures, Pre-cast Concrete, and Pre-stressed Concrete designs.
- **Vetting & Audits:** Personally vetted and approved over **500 structural design projects**.
- **Key Projects Handled:**
 - *Army School Shillong*
 - *Vidya Mandir School building complex (Aranthangi)*
 - *Madras Diabetes Research Center*
 - *Marg Ltd* infrastructure projects

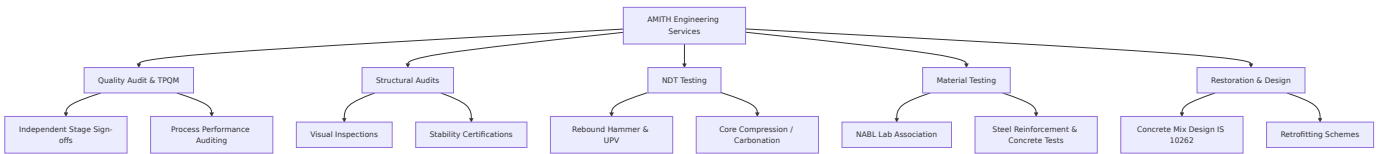
- *RCC Bridge* at Wellington, Ooty
- *Airport Authority of India (1.5 Million Liter Capacity Sump)*
- Structural audits/works at *MES Tambaram* and *MES Port Blair*.

3. Er. Kalaimony R

- **Background:** Retired Senior Engineer from **TANGEDCO** (23 years of public service) and residential developer (15 years, completed 40+ housing projects).
- **Key Role:** Senior Consultant for the **Centre for Urbanization, Buildings & Environment (CUBE)** - IIT Madras.
- **Key Projects Handled:**
 - *Andhra Pradesh Capital Region Development Authority (APCRDA):* Amaravati Secretariat and Government Housing project oversight.
 - *Tamil Nadu Housing Board (TNHB):* Third-Party Quality Monitoring (TPQM) and condition assessment of 9-storey towers at Korattur.
 - *Tamil Nadu Urban Habitat Development Board:* TPQM of residential blocks at Vallam, Thanjavur.
 - *Kendriya Vidyalaya School Projects:* Supervision of school buildings across Kerala, Karnataka, and Madhya Pradesh.

3.4 Services Page

- **Meta Title:** Quality Control Inspection Services in Chennai, TamilNadu, India
- **Detailed Services & Work Scope:**



1. Quality Audit/ Assessment & TPQM (Third-Party Quality Monitoring):

- Focuses on preventing construction defects from day one. Involves independent audits of material quality, reinforcement placement, concrete pouring, and finishing processes.

2. Structural Auditing:

- Physical, visual, and analytical testing of structures (columns, beams, slabs) to identify cracks, deflection, rebar exposure, and overall building stability.

3. NDT (Non-Destructive Testing) & Partial Destructive Testing:

- *Rebound Hammer*: Verifies surface hardness of concrete.
- *Ultrasonic Pulse Velocity (UPV)*: Checks concrete homogeneity and detects internal voids/cracks.
- *Carbonation & Chloride Tests*: Assesses carbonation depth and chemical penetration leading to rebar corrosion.
- *Concrete Core Drilling & Testing*: Extracts physical cylinders to evaluate compressive strength in lab.
- *Half-Cell Potential*: Measures corrosion probability of embedded reinforcement.
- *Chemical Analysis*: Evaluates concrete/mortar composition.

4. Testing of Building Materials:

- Compression tests on bricks, blocks, concrete cubes.
- Tensile, bend, and rebend testing of steel reinforcements and High-Tensile Steel (HTS) strands.
- Chemical analysis of water and soil (pH value, sulfates, chlorides).

5. Restoration & Rehabilitation of Structures:

- Developing structural repair schemes, specifying epoxy grouting, carbon wrap, micro-concrete jacketing, and cathode protection methods.

6. Concrete Mix Design:

- Providing scientific concrete mix proportioning matching Bureau of Indian Standards (**IS 10262** and **IS 456**) using normal or accelerated curing methods.

7. Consultancy & Advisory Services:

- Feasibility reports, design validation, structural peer reviews, and academic consultation.

8. Dissemination of Technical Knowledge:

- Conducting targeted training programs, workshops, and seminars for builders, site engineers, and government departments on quality control.

3.5 Gallery Page

- **Meta Title:** Repair and Rehabilitation of Concrete Structures Chennai, India

- **Structure:** Uses dynamic categorization:
 - *Structural and Stability audit*
 - *QAQC and TPQM*
 - *Restoration and Rehabilitation* (spelled "*Rehabilitaion*" in old codebase)

3.6 Contact Us Page

- **Form Submissions:** Captures Name, Phone, Email, and Message.
- **Footer Structure:** Quick Links mapping, local address info, and a "Get A Quote" email capture.

4. Gap Analysis: Old Site vs. Redesign Codebase

Comparing the live site (<https://amith.in.net/>) to the Next.js source code under `AMITH_REDESIGN/src/app/` , the following discrepancies and gaps have been identified:

Element / Area	Old Site State	Redesign Codebase State (AMITH_REDESIGN)	Impact / Recommendation
Director Profiles	Detailed dossiers outlining projects, years of service, academic degrees, and institutional titles.	Short 2-sentence summaries with generic roles inside WhyChooseUs.tsx .	Critical. Credibility is lost. The detailed biographies should be added to the Redesign as expandable details or modal overlays.
Services Content	In-depth descriptions of tests, laboratory integrations, and standard operating procedures.	Bullet features inside Services.tsx . Contains a Lorem Ipsum placeholder in the old website.	Medium. Make sure the text on QA/QC TPQM contains real copy, not standard filler text.

Projects Showcase	Showcases real infrastructure (Sewage plants, Slum tenements, run-ways, schools) via Director profiles.	Uses four placeholder projects with Unsplash image links in Projects.tsx .	High. Replace placeholder cards with the actual projects handled by the Directors (e.g., North Madras Power, TNHB TPQM, APCRDA Amaravati).
Gallery Page	Tabbed grid of 20+ images showing testing equipment, concrete cores, and sites.	Completely missing a dedicated gallery grid; only standard static showcase items exist.	Medium. Implement a modern, filterable grid using the categories: <i>Stability Audit</i> , <i>QAQC/TPQM</i> , and <i>Restoration/Rehab</i> .
Social Media Links	Social media icon indicators.	Icons exist but link to empty anchors # with default click prevention in Footer.tsx .	Low. Keep placeholders inactive or update to corporate profiles.
Spelling Errors	Has spelling typo: <i>"Rehabilitaion"</i> in gallery tab.	Spelled correctly in code, but verify text consistency.	Low. Fixed in Redesign components.

5. Actionable Design & Technical Enhancements

To deliver a premium, high-impact web application that outclasses the outdated site while maintaining accuracy, the following improvements are recommended:

1. Expand the Leadership Section (WhyChooseUs.tsx)

Expand the director card structures to support a modal pop-up or a toggle drawer that displays the full credentials, academic histories, and detailed projects of Dr. Devadas, Col. Nallathambi, and Er. Kalaimony.

2. Implement the Projects Gallery Component

Create a dynamic, filter-based Gallery section that matches the old categories (Stability Audit, QA/QC, Restoration). Use smooth CSS animations or Framer Motion transitions as user filters between categories.

3. SEO Integration

Port the old site's metadata directly. Define an exportable `metadata` object in the Next.js `layout.tsx` or `page.tsx` with optimized titles, description strings, and JSON-LD schema referencing the three board directors and their NABL laboratory partners.

4. Interactive NDT Testing Showcase

On the services grid, allow clients to click/hover on tests (like Rebound Hammer, UPV, Carbonation) to reveal brief micro-explanations of how the test is performed and what anomalies it detects.